



Society of Petroleum Engineers

SPE-227146-MS

Data-Driven Modeling to Estimate Wettability in Hydrogen-Brine-Mineral Systems: Insights for Underground Hydrogen Storage

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227146-MS](https://doi.org/10.2118/227146-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Underground geological structures have gained attention for large-scale hydrogen storage. However, this approach faces significant challenges due to the complex subsurface wettability behavior. Experimental techniques are commonly used to understand such complexities; however, they are tedious and expensive. In contrast, machine learning is an effective alternative to predict wettability, but existing models are limited in capturing the nuanced mineralogical interactions. This research proposed a novel model for predicting wettability in Hydrogen-Brine-Mineral Systems.

An experimental dataset was collected from the literature, consisting of hydrogen/brine contact angles on pure clean surfaces, under various pressures (10 - 100 bar), temperatures (20 - 80 °C), and salinity conditions (0 - 122347 ppm). Three machine learning models-Support Vector Regression (SVR), Gradient Boosting Regression (GBR), and Extreme Gradient Boosting (XGBoost) were employed to predict the contact angles. The models were trained and validated using KFold cross-validation, and the hyperparameters were optimized through an exhaustive grid-search approach. The model performance was evaluated using the mean absolute percentage error (MAPE), root mean squared error (RMSE), and coefficient of determination (R^2).

Data analytics illustrate that the contact angle in such systems is highly dependent on the mineral surface, followed by pressure, temperature, and salinity. The comparative performance analysis of the applied models showed that XGBoost outperformed the others, achieving MAPE of 1.13% (training) and 5.67% (testing) with R^2 of 98 % on the testing data. The proposed model can predict contact angles for Hydrogen-Brine-Mineral Systems as a function of pressure, temperature, mineral specific gravity, and salinity. Different types of minerals considered in this study include: calcite, dolomite, quartz, granite, anhydrite, gypsum, shale, and basalt. The trained model was tested on an unseen dataset. A group trend analysis was also conducted to confirm the model's integrity. Contact angles generally increase with both pressure and temperature due to changes in hydrogen density and interfacial interactions, with pressure having a more pronounced impact. Mineral composition significantly influences wettability; for instance, gypsum and anhydrite exhibit higher contact angles than basalt and granite under the same conditions. Increased salinity also led to higher contact angles due to ion-induced modification of the surface properties.

A novel machine learning model was proposed in this study to predict contact angles for Hydrogen-Brine-Mineral systems. The proposed model may save a significant amount of time when running tedious experiments and molecular simulations, thus adding value to the literature.

Keywords: Contact Angle, Hydrogen geo-storage, Machine Learning, Rock Minerals, Wettability

Introduction

The global transition toward a low-carbon energy future has raised the interest in hydrogen as a clean and versatile energy carrier. Hydrogen's high energy density and potential for emission-free utilization make it an attractive candidate for decarbonizing transportation, industry, and power supply (Akpasi et al., 2025; Kalam et al., 2023). However, the intermittent nature of renewable energy sources requires efficient and high-capacity energy storage solutions that can match the supply and demand. Underground hydrogen storage (UHS) in geological structures has become a promising option for storing excess renewable energy in the form of hydrogen, thereby facilitating energy security and the stability of power grids (Bade et al., 2024; Gomez et al., 2024; Taiwo et al., 2024).

Geologic structures like depleted oil and gas reservoirs, aquifers, and salt caverns are large storage formations for hydrogen. These structures are well studied for storing natural gas and form a solid ground for modifying them for hydrogen storage applications (Bade et al., 2024; Taiwo et al., 2024). Hydrogen, being a gas with low molecular weight, high diffusivity, and reactivity, presents different and specific storage challenges in subsurface environments (Miocic et al., 2023). It is important to understand the behavior and interactions of hydrogen, brine, and rock minerals for evaluation of storage performance, safety, and efficiency.

One of the key factors influencing hydrogen behavior in geological formations is wettability, which describes the preference of a solid surface to be in contact with one fluid over another (Al-Mukainah et al., 2022; Hosseini et al., 2022). Wettability affects the fluid distribution, flow, and trapping mechanisms within porous media, affecting the injectivity, deliverability, and retention of stored hydrogen (Higgs et al., 2022; Pan et al., 2021). The contact angle, defined as the angle at which a fluid interface meets a solid surface, serves as a quantitative measure of wettability (Hashemi et al., 2021). A low contact angle indicates a water-wet system, in which brine preferentially wets the rock surface, whereas a high contact angle suggests a hydrogen-wet system, favoring hydrogen adhesion. Numerous studies have investigated how pressure, temperature, and salinity affect wettability in hydrogen-brine-rock systems (Ali et al., 2025; Dehghani et al., 2024; Esfandyari et al., 2022; Janjua et al., 2024). For example, some studies have demonstrated that pressure and temperature variations can change the contact angle, leading to the alteration of the wettability behavior of the system. Other factors influencing the complexity of wettability behavior include the mineralogical constitution of the rock, roughness of the surface, and fluid-solid interface chemistry (Alanazi et al., 2023; Ali et al., 2025; Al-Yaseri & Jha, 2021; Esfandyari et al., 2024; Hashemi et al., 2021; Hosseini et al., 2022; Hosseini et al., 2022; Iglaue et al., 2021). These factors all point toward developing robust models capable of predicting wettability under a range of subsurface conditions.

Improvements in machine learning (ML) have made it possible to simulate more complicated underground processes. ML algorithms can use large datasets to search for connections between different variables and provide predictions that complement traditional laboratory and numerical methods. By using ML models, it is now possible to predict interfacial tension and wettability in hydrogen storage, which helps with the assessment and optimization of storage sites (Ali et al., 2024; Hosseini & Leonenko, 2024; Maleki et al., 2024; Vo Thanh et al., 2023).

The current state of ML in this area mainly focuses on one aspect of the hydrogen-brine-rock system or works within certain limits of minerals and conditions. No fully developed models exist that use pressure, temperature, salinity, and mineralogy together to predict contact angles in various subsurface conditions. Being aware of this research gap allows us to advance wettability knowledge in hydrogen storage and to

make trusted tools for use in selection and operational design. We aim to develop and train ML models to predict contact angles in hydrogen-brine-rock system based on important geological and operational parameters. In this study, we seek to use experimental data and advanced ML techniques to make wettability models more accurate and useful, which will allow for the safe and effective operation of underground hydrogen storage sites.

Methodology

Data Description and Analytics

Data analysis was performed as a fundamental step to ensure the reliability and interpretability of the machine learning models developed to predict contact angles in hydrogen-brine-mineral systems. A dataset comprising 256 experimental data points was compiled from the literature (Esfandyari et al., 2022). The experimental measurements included hydrogen/brine contact angles on eight clean mineral surfaces (calcite, dolomite, quartz, granite, anhydrite, gypsum, shale and basalt). All samples were aged in an organic acid solution (10^{-2} mol/L stearic acid) for 7 days at an elevated temperature of about 60 °C and atmospheric pressure to mimic subsurface conditions. The measurements covered a variety of pressure (10 - 100 bar), temperature (20 - 80 °C), and salinity (0 - 122347 ppm) conditions. It is worth mentioning that this work does not consider the effect of the cushion gases (CO_2 , CH_4 , or N_2) on the hydrogen–brine contact angle. A statistical summary of the dataset, shown in Table 1, was developed to assess the variability required to train the machine learning models and the major tendencies across all input and output parameters.

Table 1—Statistical analysis of the training dataset

	Temperature (°C)	Pressure (bar)	Salinity (ppm)	Mineral Type	Contact Angle (°)
Count	204	204	204	204	204
Minimum	20	10	0	1	15
Maximum	80	100	122347	8	93
Mean	50.20	56.18	59974.02	4.53	43.95
Standard Deviation	22.68	33.47	61312.20	2.32	14.73
25 percent	20	40	0	3	34.75
50 percent	60	70	0	4.5	43.5
75 percent	80	100	122347	7	53

The dataset was split into training and testing sets at an 80:20 ratio. As part of the initial exploration of the data, a Pearson correlation matrix was created and visualized as a heat map (Figure 1) to assess the linear relationships between the input features and the target variable, namely, the contact angle. The analysis shows that temperature has the most significant positive correlation with contact angle, followed by pressure and salinity, suggesting that these variables play a significant role in wettability alterations. Additionally, the low correlation values among the input parameters themselves indicate that they are largely independent, which is favorable for machine learning modeling as it improves model reliability.

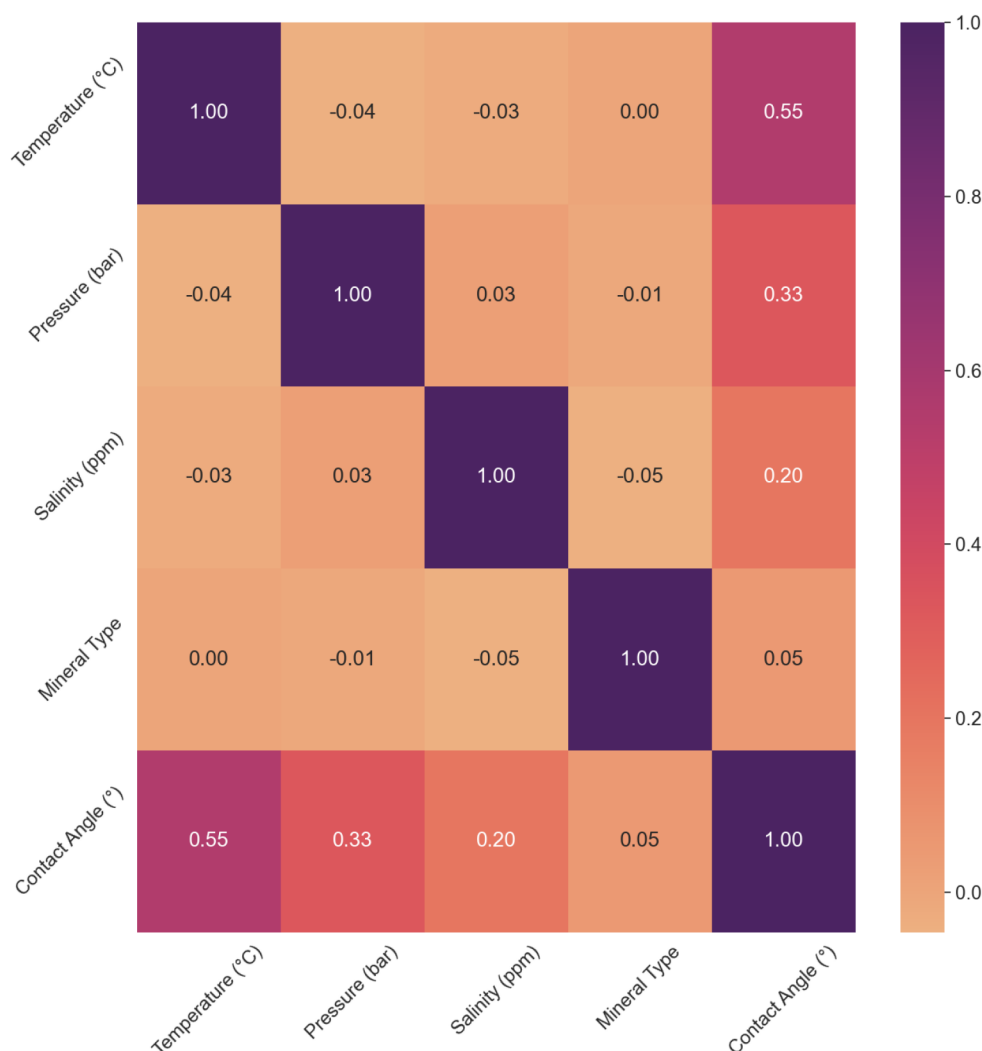


Figure 1—Correlation coefficient heat map of the training dataset

Figure 2 shows a pair plot showing how temperature, pressure, salinity, and mineral type are related to each other. This was done to learn more about how the input variables are related to each other. This visual tool helped identify patterns and groupings in the dataset. For instance, minerals are shown as separate vertical layers because they are categorized, whereas pressure and temperature vary more smoothly. In contrast, salinity forms clear clusters representing the different levels used in the experiments. The diagonal histograms provide a quick sense of how the data is distributed for each variable. The color intensity in the plots indicates the density or frequency of data points, with darker shades representing higher concentrations. Overall, the pair plot provides a clear picture of how these parameters are connected and guides the impact of the contact angle predictions during the modeling process.

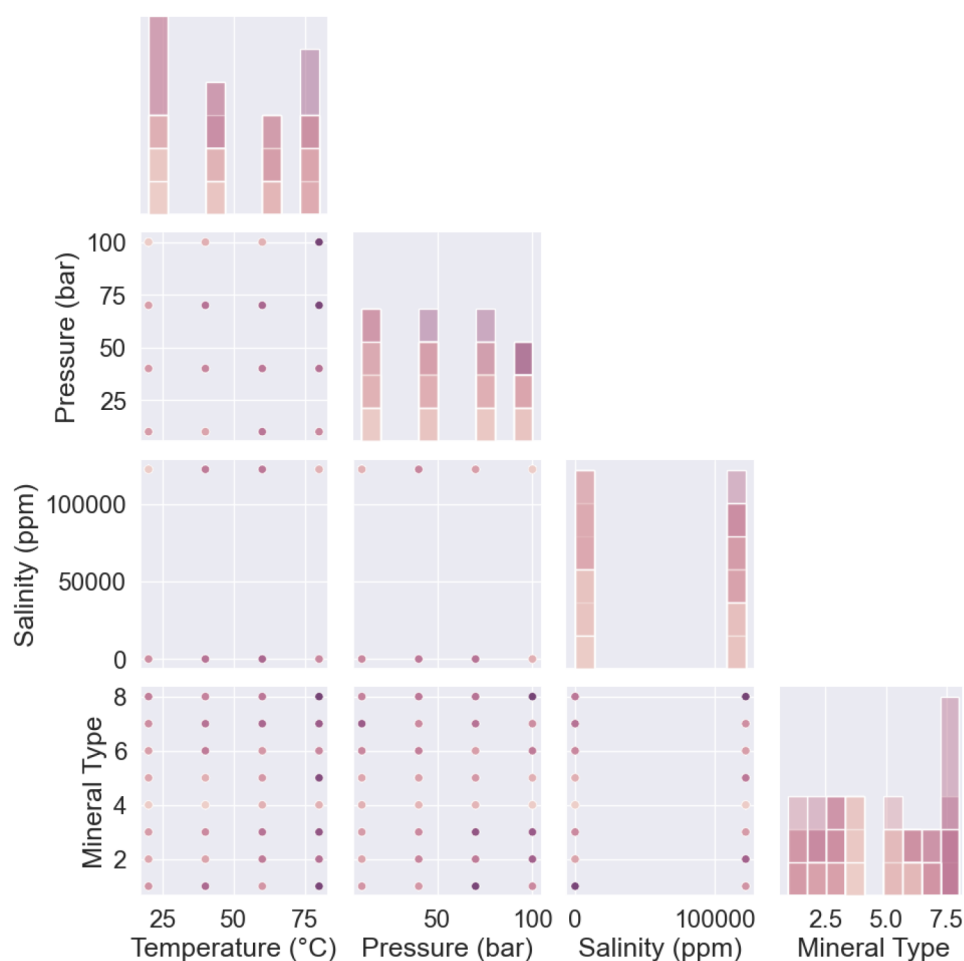


Figure 2—Pair plot of the training dataset

Modeling Tools

Three ML models were used in this research: Extreme Gradient Boosting (XGBoost), Gradient Boosting Regression (GBR) and Support Vector Regression (SVR). Each model was selected for its particular method for learning the learning underlying patterns from the data. All models were trained and tested on an identical dataset to facilitate an even comparison. The aim of this study was not only to examine their predictability but also to determine how their respective architecture affects performance when trained on complicated, nonlinear systems such as hydrogen-brine-mineral interactions.

Figure 3 illustrates the workflow. Model training and hyperparameter tuning were iterated to enhance prediction accuracy. A combined grid search and 5-fold cross-validation approach was used. In this method, the data are divided into five folds, with each subset serving as a validation set while the remaining folds train the model. The grid search evaluates parameter combinations to identify the optimal configuration. This approach mitigated overfitting and improved model robustness.

In assessing the performance of the machine learning models, three statistical metrics were utilized: Mean Absolute Percentage Error (MAPE), coefficient of determination (R^2), and Root Mean Square Error (RMSE). MAPE computes the average absolute error in relation to the target values and is a scale-independent measure, specifically well-suited for comparison of models between different datasets. R^2 indicates how well the predicted values match the true data, meaning that values closer to 1 represent a better performing model and stronger fit. The RMSE computes the square root of the average of the squared prediction and true value differences, giving more prominence to larger differences. The mathematical equations for MAPE, R^2 and RMSE are shown in Equations (1) to (3) below:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{A_i - P_i}{A_i} \right| \times 100 \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (A_i - P_i)^2}{\sum_{i=1}^n (A_i - \bar{A}_i)^2} \quad (2)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (A_i - P_i)^2}{n}} \quad (3)$$

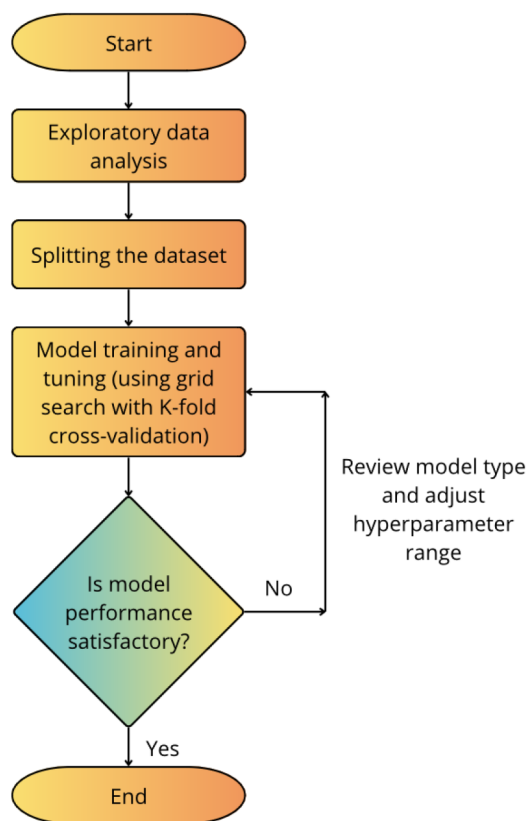


Figure 3—Workflow diagram of machine learning model construction

XGBoost. Extreme Gradient Boosting (XGBoost) is a fast and scalable version of gradient boosting algorithms that was developed to improve speed and performance. It constructs an ensemble of decision trees sequentially, and each tree attempts to correct the errors of its previous one by optimizing a regularized objective function. XGBoost has several innovations, such as regularization to avoid overfitting, parallelized tree construction, and treating missing values, that are particularly effective in modeling nonlinear and high-dimensional datasets (Chen & Guestrin, 2016).

GBR. Gradient Boosting Regression (GBR) is an ensemble learning method that combines the prediction of several weak learners, often decision trees, to generate a strong prediction model. In contrast to random forests, which construct trees separately, trees in GBR are built sequentially, with every successive new tree being tasked with optimizing the residual errors of the collective previous trees through the use of gradient descent optimization. By iteratively refining, the model can learn intricate patterns in the data. Although it is extremely accurate, the GBR is highly susceptible to overfitting and tuning of the hyperparameters; thus, cross-validation and regularization are necessary for generalization (Friedman, 2001; Kalam et al., 2023)

SVR. Support Vector Regression (SVR) builds a hyperplane with the aim of best fitting the data within a given margin of tolerance or epsilon. SVR utilizes the use of kernel functions, such as the radial

basis function (RBF), for transforming input data into a higher dimension in which linear separability is achievable. SVR, by considering only those data points located outside the epsilon margin, offers a sparse and robust regression system, which proves to work best in cases of small datasets or in situations in which the underlying patterns are highly nonlinear (Kalam et al., 2024; Smola & Schölkopf, 2004).

Results and Discussion

Hyperparameters Tuning

XGBoost. XGBoost was used to model contact angles in the hydrogen-brine-mineral system with hyperparameters tuned via a grid search (Table 2). The process of tuning covered a large set of values for a number of critical parameters, such as the number of estimators (25–2000), maximum depth (2–10), and learning rate (0.01 to 0.3). The optimum configuration comprised 2000 estimators, a depth of 4, and a learning rate equal to 0.2, which together preserved the learning speed against overfitting and generalization. Regularization was achieved by setting the minimum child weight to 10 and leaving default values for gamma, alpha, and lambda, with only the lambda (set at 1) directly contributing to model stability. The subsampling of rows and columns as well as feature sub-sampling by tree hyperparameters was set to 1 to ensure that the model captured the entire complexity of the data. This optimized configuration allowed XGBoost to successfully model the nonlinear relationships between the input variables and contact angle, making it a robust choice for this predictive task.

Table 2—Summary of hyperparameters used for the three ML models in this study

ML Tool	Hyperparameter	Range	Optimized value
XGBoost	Number of estimators	25 - 2000	2000
	Max depth	2 - 10	4
	Learning rate	0.01 - 0.3	0.2
	Min child weight	2 - 10	10
	Gamma	0	0
	Subsample	1	1
	Column subsample by tree	1	1
	Alpha	0	0
	Lambda	1	1
GBR	Number of trees	25 - 2000	1000
	Max features	sqrt, log2, auto	sqrt
	Max depth	2 - 10	4
	Min samples were split	2 - 10	6
	Min sample leaf	1	1
SVR	Kernel	linear, poly, rf, sigmoid,	rf
	C	0.1, 1, 10, 100, and 1000	100
	Epsilon	1, 0.1, 10 ⁻³ , 10 ⁻⁵ , 10 ⁻⁷ , 10 ⁻⁹	1
	Gamma	auto, scale	scale

GBR. The GBR was also used to simulate contact angle behavior, with its most critical hyperparameters optimized to maximize predictive accuracy. A grid search was performed for the number of trees, feature selection method, maximum depth, and minimum sample requirements for splitting and leaf creation (Table

2). The boosting iterations number was experimented with from 25 to 2000, with 1000 yielding the best accuracy. The model was optimal with maximum depth set as 4 so that it could learn nonlinearity without overfitting. Sqrt was the best setting for the maximum number of features. The minimum splitting sample was set to 6, so that the split is only permitted when there are the required samples to allow for it, while the minimum sample for the leaf was set to 1, to allow for flexibility in the creation of the leaves. These combinations of hyperparameters resulted in a well-tuned GBR model that was effective in learning from the data without losing generalization to unseen samples.

SVR. In the SVR model, a series of kernel functions including linear, polynomial, radial basis function (RBF), and sigmoid functions was used, with the RBF kernel finally chosen for the best performance in resolving complex feature interactions. The C regularization parameter was varied over a broad interval (0.1 to 1000), with the optimum performance at C being 100. The epsilon parameter was varied across orders of magnitude, with epsilon = 1 giving the best results by providing flexibility in following the underlying trend with robustness. For the gamma coefficient, we selected a scaling setting that automatically varied depending on the variance of the features. The hyperparameters with the values for optimum ML performance are given in Table 2.

Performance Comparison

A comparative performance evaluation was performed between the three machine learning algorithms: XGBoost, GBR and SVR. The analysis was completed with respect to the most critical evaluation metrics MAPE, RMSE, and R^2 , followed by a comparative scatter plot analysis to determine the predictive accuracy. As summarized in Table 3, both XGBoost and GBR demonstrated outstanding performance during training with R^2 values of 0.99 and low RMSE values of 0.55 and 0.51, respectively. However, SVR recorded an apparent higher training error with MAPE values of 11.19 and an overall lower R^2 of 0.81. When evaluated with the test data, both GBR and XGBoost retained their robustness, with XGBoost performing slightly better than GBR, with a lower MAPE (5.67) and higher R^2 (0.98) compared to GBR (MAPE = 6.24, R^2 = 0.97). By contrast, the SVR recorded a dramatic performance decline (R^2 = 0.68 and MAPE = 22.66), which also highlights the limitations concerning generalization.

Table 3—Performance comparison of applied three ML tools

		XGBoost	GBR	SVR
Training Data	MAPE, -	1.13	0.98	11.19
	RMSE, °	0.55	0.51	6.37
	R^2 , -	0.99	0.99	0.81
Testing Data	MAPE, -	5.67	6.24	22.66
	RMSE, °	2.59	2.80	9.43
	R^2 , -	0.98	0.97	0.68

The comparative scatter plots for both training and testing datasets, shown in Figures 4 and 5, also support the above findings, and points predicted by GBR and XGBoost lie together clustered well near the 45° line of reference, reflecting precise predictions. Data points of GBR were overlapped by ones resulted from XGBoost. For SVR predictions, the deviation and dispersion are larger, especially at the middle to higher contact angles. To realize fast and reproducible deployment, the final version of the XGBoost model was exported and saved as a Python object (.pkl file). Overall, tree-based models, particularly XGBoost and GBR, proved to be the most effective for estimating hydrogen wettability, offering a reliable alternative to time-consuming experimental methods.

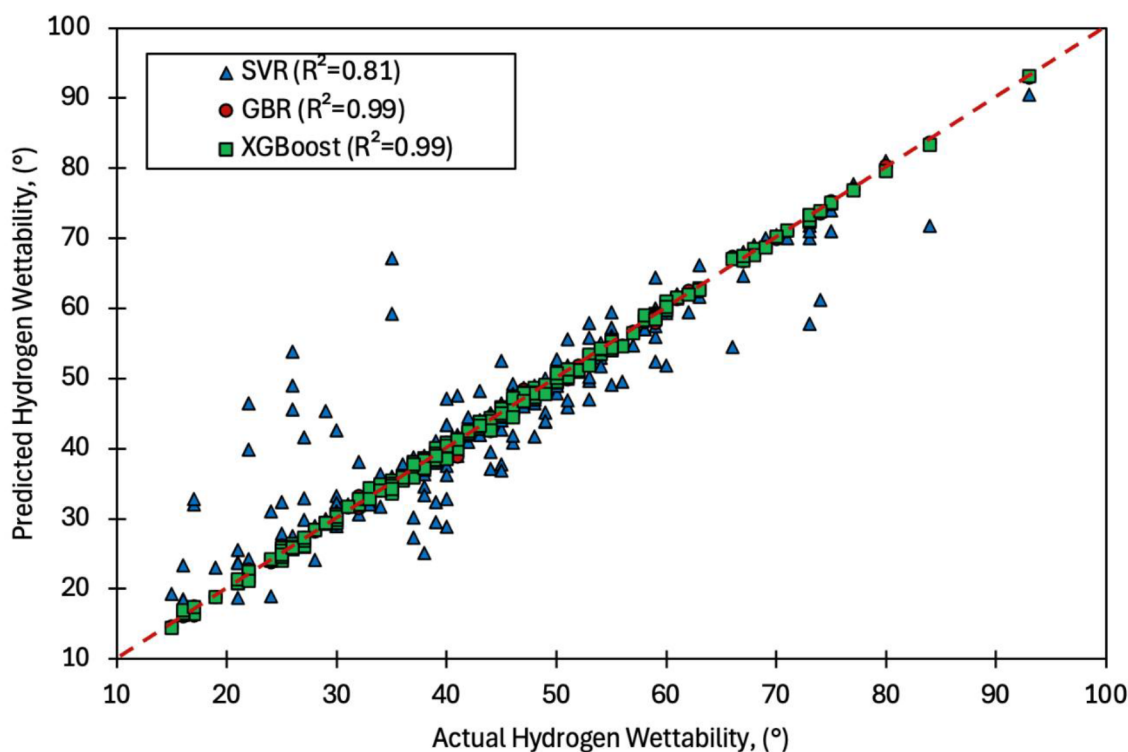


Figure 4—Scatter plots of the training datasets of the three ML tools

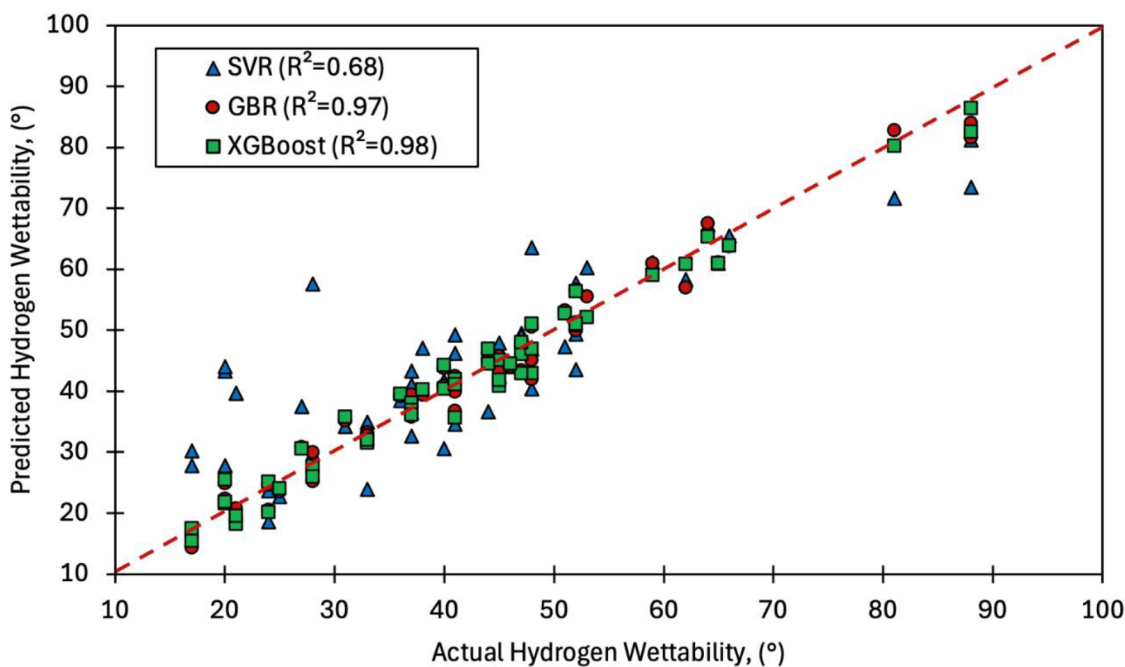


Figure 5—Scatter plots of the testing datasets of the three ML tools

Trend Analysis

Effects of Pressure and Temperature. Figure 6 shows the influence of temperature and pressure on the hydrogen/brine contact angles for calcite surfaces using distilled water, comparing experimental measurements with predictions from the XGBoost model. The data show that the contact angle increases with increasing temperature across all pressure values, indicating enhanced hydrogen wettability. This trend aligns with findings from previous studies, which have reported that higher temperatures can reduce

interfacial tension (IFT), thereby increasing contact angles in hydrogen-brine systems (Ali et al., 2025; Dehghani et al., 2024; Ibrahim, 2025; Janjua et al., 2024; Lv et al., 2024). However, the effect of pressure is even more pronounced, so that at any fixed temperature, an increase in pressure consistently leads to higher contact angles, suggesting that elevated pressures may enhance hydrogen adsorption or densification at the mineral surface (Esfandyari et al., 2022, 2024). Among the tested pressures, the difference in contact angle between 10 bar and 100 bar at 80 °C, for instance, is significantly larger than that induced by a temperature rise at fixed pressure. Additionally, the pressure-induced increase in contact angle becomes more pronounced at higher temperatures, for instance, the contact angle rise from 10 to 100 bar is steeper at 80 °C than at 20 °C. This suggests a synergistic interaction, where an increase in temperature enhances compressibility of hydrogen as well as brine displacement, increasing pressure effects on wettability. Predictions from the XGBoost model closely follow experimental values, especially under pressures of 10 and 100 bar, reflecting its ability to accurately capture synergistic effects between temperature and pressure on wettability. Low and uniform deviation between predicted and observed values under test conditions reflects on the model's consistency in predicting wettability response under different subsurface conditions.

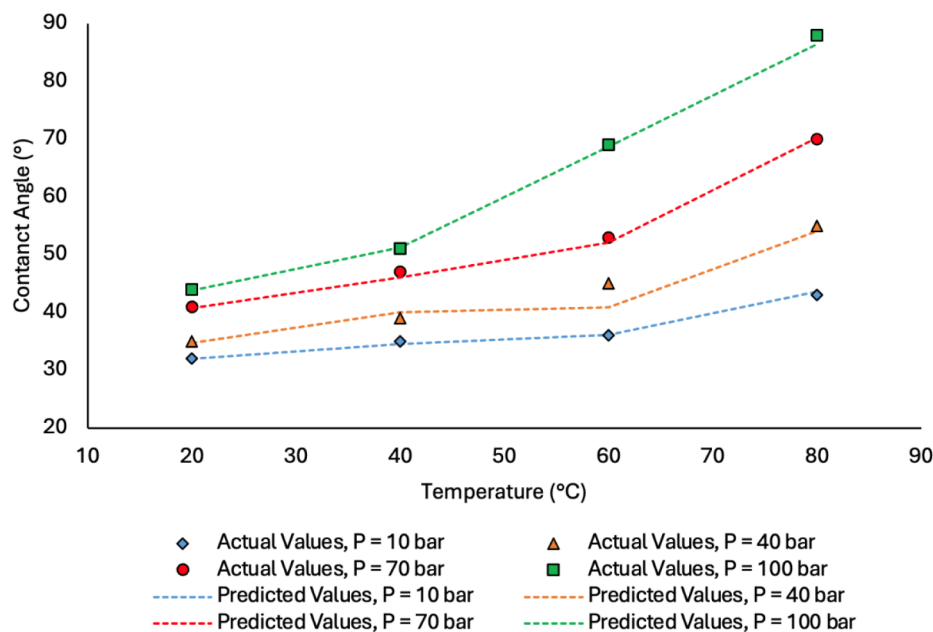


Figure 6—XGBoost-based comparison of actual and predicted hydrogen/brine contact angles on calcite under varying pressure and temperature conditions using distilled water

Effects of Minerals. Figure 7 shows the impact of mineral type on the hydrogen/brine contact angle at 40 °C and comparison of measured and predicted XGBoost measurements. The results show a strong dependency of the wettability on the mineralogical composition and surface characteristics. Among the minerals tested, gypsum and anhydrite showed the highest contact angles throughout the pressure range, indicating a more hydrophobic surface preference under hydrogen/brine conditions (Esfandyari et al., 2022, 2024). Conversely, basalt and granite showed the lowest contact angles, indicating a more water-wet tendency due to the mixing of their hydrophilic and hydrophobic phases being mixed (Alanazi et al., 2023; Ali et al., 2024; Al-Yaseri & Jha, 2021; Esfandyari et al., 2024; Hosseini et al., 2022). Quartz, dolomite, shale, and calcite showed intermediate responses, which are reflection of differences in the surface energy, roughness, and chemical composition of individual minerals (Esfandyari et al., 2022; Hashemi et al., 2021; Hosseini et al., 2022; Iglauder et al., 2021). These trends highlight the mineral-dependent wettability behavior of subsurface hydrogen systems. Increasing pressure caused the contact angles to rise for all minerals, albeit the magnitude of rise showed variability, with more reactive or porous minerals exhibiting larger

sensitivity (Ali et al., 2025; Esfandiyari et al., 2022, 2024). The XGBoost model captures these mineral-dependent trends accurately and predicted curves closely following the experimentally measured ones. The fact that the model effectively accounts for the complex effects of minerals on wettability, which is critical for interpreting hydrogen migration, trapping, and storage behavior in heterogeneous geological units, is well demonstrated here.

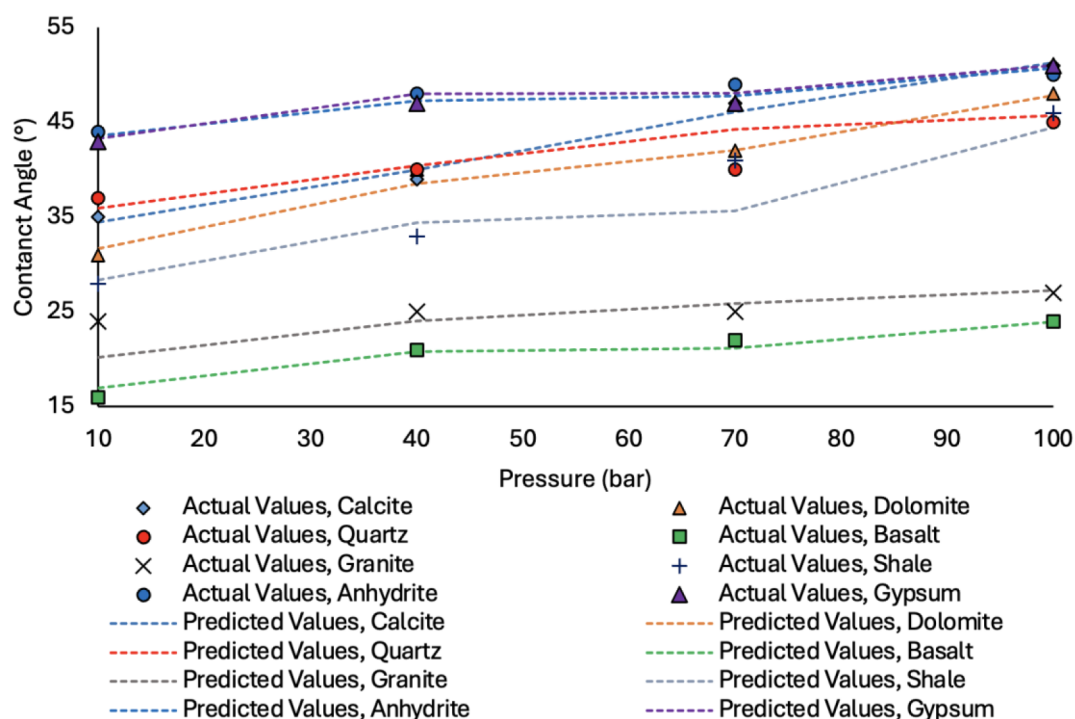


Figure 7—Comparison of experimental and XGBoost-predicted hydrogen/brine contact angles across different mineral types in distilled water at 40 °C

Effect of Salinity. Figure 8 shows the effect of salinity on the hydrogen/brine contact angles for calcite surfaces at 40 °C and a comparison between experimental measurements and the predicted XGBoost values. The output reveals that contact angles increase with increasing salinity throughout the pressure range, showing a shift toward more hydrophobic surface behavior. This effect is caused by ion-induced surface modifications at the solid/liquid boundary, where the surface charge distributions are altered and brine adhesion to the surface is reduced by the increased ionic strength. These effects have been reported in the literature where high salinity increased hydrogen/brine IFT and contact angles, reducing brine wettability and causing the surface of the rock to become more hydrogen-wet (Ali et al., 2025; Dehghani et al., 2024; Esfandiyari et al., 2022; Hosseini et al., 2022; Janjua et al., 2024). These salinity-dependent trends were well captured by the XGBoost model, and the predictions were in close agreement with the measured values. The consistency of the model in simulating the wettability behavior under different salinity conditions is important for predicting the efficiency of hydrogen storage in subsurface environments.

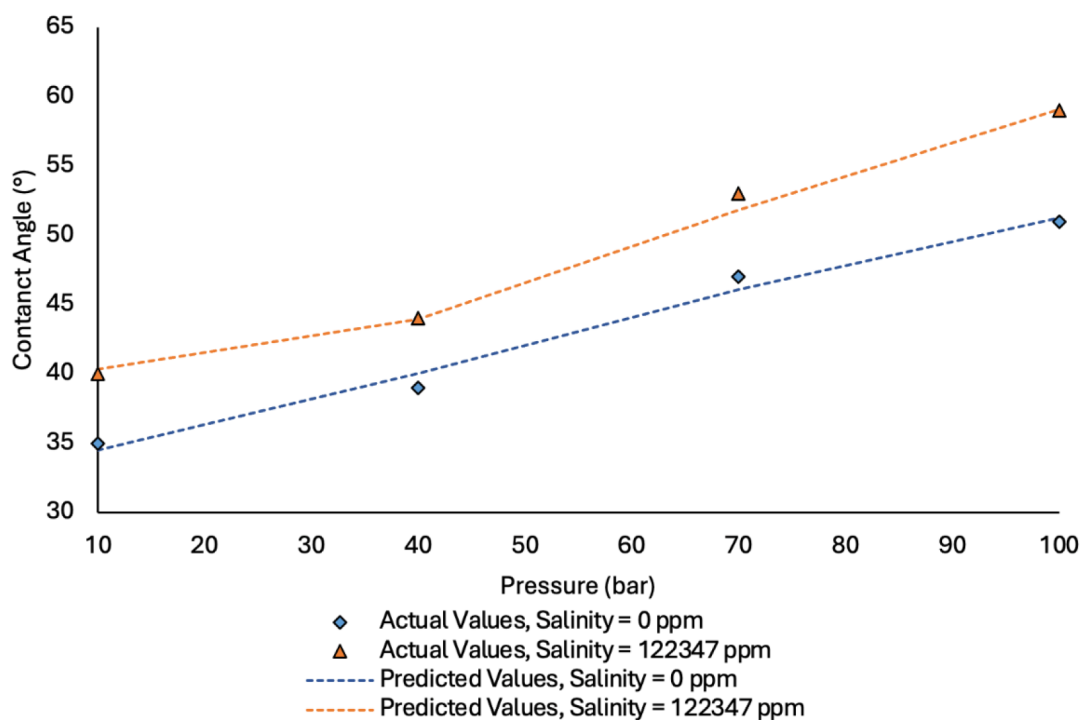


Figure 8—Comparison of measured and XGBoost-predicted contact angles for calcite in a hydrogen/brine system at 40 °C under varying salinity conditions

Conclusions

This work provides a data-driven modeling framework for predicting wettability in hydrogen-brine-mineral systems as a critical consideration for the efficiency and feasibility of underground hydrogen storage. Based on published research study's experimental data, we assembled a large dataset spanning a broad range of geological and operation conditions, including pressures ranging from 10 to 100 bar, temperatures from 20 to 80 °C, salinities up to 122347 ppm, and eight common types of minerals in subsurface formations. Three machine learning models (XGBoost, GBR, and SVR) were trained on 5-fold cross-validation and hyperparameter tuning through grid search. As a result, the XGBoost model outperformed the other models, with MAPE of 1.13 and R^2 of 99% for training and MAPE of 5.67 and R^2 of 98% for testing.

The feature trend analysis revealed that the pressure had the most notable effect on the contact angle, likely due to hydrogen densification and enhanced surface adsorption at higher pressures. For instance, on calcite surfaces at 80 °C, the contact angle increased from 43° at 10 bar to 88° at 100 bar, highlighting the magnitude of pressure influence. Increased salinity increased contact angles, proving that ion-induced effects diminish brine adhesion to solid surfaces. At a constant pressure and temperature, the contact angle on calcite increased from 51° with 0 ppm salinity to 59° with 122347 ppm salinity, illustrating the impact of ionic strength. Mineralogical effects were also evaluated, indicating that gypsum and anhydrite exhibited hydrophobic behavior (higher contact angles) and granite and basalt showing water-wet characteristics. These findings are in agreement with previous laboratory work.

The final XGBoost model was saved in a.pkl file, offering a practical tool for the fast prediction of contact angle values for a broad range of UHS conditions. This demonstrates the potential for applications of machine learning, especially tree-based ensemble models, in complementing or substituting time-consuming and expensive laboratory and simulation-based approaches.

Future studies would involve broadening the dataset to include time-dependent effects, other rock types, and more extensive pressure and temperature conditions. It will be an important contributor to digital workflows in subsurface energy storage applications.

Nomenclature

Symbol	Description	Unit
A_i	Actual value at data point	degrees (°)
P_i	Predicted value at data point degrees	degrees (°)
$\bar{A}_i$	Mean of actual values	degrees (°)
n	Total number of data points	-
MAPE	Mean Absolute Percentage Error	-
RMSE	Root Mean Squared Error	degrees (°)
R^2	Coefficient of determination	-

Acknowledgment

The authors would like to acknowledge the support of Nazarbayev University through the NU Faculty Development Competitive Research Grants Program (Grant number: 201223FD2608).

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